

Defense / Autonomy

Tracking, radar image formation, and GPS-denied navigation.

THREE SYSTEMS TO INSPECT

01

track-fusion

Python

47%

LOCALIZATION ERROR REDUCTION

67 tests; IMM coverage rises from 0.893 to 0.960, while the published sweep shows where the bank loses.

FAILURE BOUNDARY

The useful IMM band is 3-6 degrees per second; at 0 and 12 degrees per second the configured bank loses.

START `git clone https://github.com/asp53826/track-fusion && cd track-fusion && python -m pip inst...`

02

sar-focus

Python

<0.5%

RESOLUTION ERROR VS THEORY

27 tests; the theory oracle exposed interpolation that produced physically impossible sidelobes.

FAILURE BOUNDARY

Synthetic phase history with idealized propagation and no measured sensor calibration errors.

START `git clone https://github.com/asp53826/sar-focus && cd sar-focus && python -m pip install -e...`

03

vio-nav

Python

51.9x

LOWER DRIFT AT 40 SECONDS

37 tests; MSCKF stays near four centimeters while dead reckoning reaches 2.106 meters.

FAILURE BOUNDARY

At five seconds VIO is worse than inertial-only, and hover makes it 13x worse because depth is unobservable.

START `git clone https://github.com/asp53826/vio-nav && cd vio-nav && python -m pip install -e '....'`

VERIFICATION CONTRACT

- Claims name the comparison baseline and units.
- Commands reproduce the public correctness path.
- Failure boundaries remain beside the measured result.
- The evidence manifest connects claims to methods and sources.